CISS245: Advanced Programming
Quiz q3001

Name: eagraham1@cougars.ccis.edu Score:

Q1. Write a struct for `Martian` so that a `Martian` variable has

- `num_heads`: number of heads
- `num_arms`: number of arms
- `num_legs`: number of legs
- `gender`: gender (as a `char`)
- `max_speed`: maximum speed of travel (as a `double`)

and create struct variable `marvin` with 1 head, 2 arms, 3 legs, and gender of `'r'` using an initializer list.
Answer:

```
// create Martian struct here
Struct Martian
{
  int num_heads;
  int num_arms;
  int num_legs;
  char gender;
  double max_speed;
}
// create marvin here
Martian marvin = {1, 2, 3, 'r', 0.0};
```

Q2. You are given

```
struct Position
{
    int x, y;
};
struct Alien
{
    Position p;
    int health;
};
```

Complete the following to create an array `x` of 100 `Aliens`, all at the given position `p` (see below) and all with `health` of 1000.
Answer:

```
Position p = {10, 20};
```

```
// create x here
Alien x[100];

for (int i = 0; i < 100; ++i)
{
  x[i].p = p;
  x[i].health = 1000;
}
```

Q3. An `Alien` struct value has a `health` member variable. Something is not quite right with the following. Fix it so that it works.
Answer:

```
Alien * alien = new Alien;
alien->health = 0;
```

Q4. Complete the following (see STEP 1 and STEP 2)) so that the program prints

```
<Weapon type:sword, name:guthwine>
<Person name:eomer, health:1000, weapon:<Weapon type:sword, name:guthwine>>
```

Answer:

```
#include <iostream>

struct Weapon
{
    char type[1024];
    char name[1024];
};

struct Person
{
    char name[1024];
    int health;
    Weapon * weapon;
};

void Weapon_print(Weapon & w)
{
    std::cout << "<Weapon type:" << w.type << ", name:" << w.name << '>';
}

void Weapon_println(Weapon & w)
{
    Weapon_print(w);
    std::cout << '\n';
}
```

```
void Person_println(Person & p)
{
    std::cout << "<Person name:" << p.name
              << ", health:" << p.health
              << ", weapon:";
    // STEP 2: Add a statement here to print the weapon of p
    Weapon_print(*p.weapon);
    std::cout << ">\n";
}

int main()
{
    Weapon w = {"sword", "guthwine"};
    Weapon_println(w); // prints <Weapon type:sword, name:guthwine>

    // STEP 1: Create variable p of type Person and with name of "eomer",
    // health of 1000, and has weapon w (from above).
    Person p = {``eomer'', 1000, &w};

    // The following should print
    // <Person name:eomer, health:1000, weapon:<Weapon type:sword, name:guthwine>>
    Person_println(p);

    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.